

The two-dimensional trichotomy:

$$Z(n) = O\left(n^{-\min(p_1, p_2)/2} \log n\right) \text{ for every } (k, h)$$

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Abstract

The general two-dimensional chart integral is symmetric under exchanging its coordinates (Fubini on the compact box). This converts unit 46's $p_1 < p_2$ theorem into the $p_1 > p_2$ one and, with unit 45's equal case, yields the unconditional leading-term statement $Z(n) = O(n^{-\min(p_1, p_2)/2} \log n)$ for arbitrary exponents $k = (k_1, k_2)$, $h = (h_1, h_2)$: the two-dimensional content of `thm:TaylorTree`'s leading term, with the logarithm present exactly when the two candidate exponents coincide. Zero `sorry/axiom`.

1 Setting

Unit 44's exact reduction singles out the first coordinate ($n^{-p_1/2}$ in front). When $p_1 > p_2$ the reduction is still exact but its integral grows like $L^{p_1-p_2}$, so the useful statement is obtained by swapping the roles of the coordinates first.

2 The things formalised

```
theorem twoDGeneral_swap ( a b n : ) (h h k k : ) :
  twoDGeneral a b n h h k k = twoDGeneral a b n h h k k
theorem twoDGeneral_isEquivalent_of_gt ...
  (hgt : ((h : ) + 1) / k < ((h : ) + 1) / k) :
  (fun n => twoDGeneral a b n h h k k) ~[atTop]
    fun n => 1 / ((k : ) * k) * b ^ (...) * noLogConst a (...) (...)
      * n ^ (-(((h : ) + 1) / k / 2))
theorem twoDGeneral_isBigO_min ... :
  (fun n => twoDGeneral a b n h h k k) =0[atTop]
    fun n => n ^ (- (min (((h : ) + 1) / k) (((h : ) + 1) / k) / 2)) * Real.log n
```

3 Proof strategy

The swap is `integral_integral_swap` for the product of the two restricted measures; the integrand is continuous on $\mathbb{R}^2$, hence integrable on the compact $[0, b]^2$ and on the box $(0, b]^2$ by monotonicity, and the two integrands agree pointwise after commuting the monomials. The trichotomy is `lt_trichotomy` on p_1, p_2 ; in each branch the `IsEquivalent` gives an `IsBigO` to its own right-hand side, which is then compared to $n^{-\min/2} \log n$ using $n^r = O(n^r \log n)$ and `IsBigO.const_mul_left`.

4 Roadblocks and resolutions

- None in Lean; the file compiled on the first pass. The line-length gate caught prose lines in the module docstring after an initial commit, which was amended.

5 What was Mathlib, what was new

Mathlib: `integral_integral_swap`, `Measure.prod_restrict`,
`ContinuousOn.integrableOn_compact`, `lt_trichotomy`, `IsBigO.const_mul_left`. New:
the symmetry and the assembled trichotomy.

6 Lessons

Once each regime has its own `IsEquivalent`, the unconditional big- O statement is a short case split; it is worth stating because it is the form in which the paper's theorem is quoted.

7 Outcome

Units 44–47 complete the two-dimensional leading-term theory of the standard integral for arbitrary (k, h) : exact reduction, three regimes with sharp constants, and the unified big- O bound. Remaining directions: mixed multiplicity in general dimension, and next-order terms in the general equal case.